

WOOSEOP HWANG

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Education

University of Oxford

Master of Mathematics and Theoretical Physics (MMathPhys)

- First Class Honours, with Distinction.

Oct 2020 – Jun 2024

Oxfordshire, United Kingdom

Experience

Qraft Technologies

Quantitative & AI Researcher (Alternative Military Service)

Aug 2024 – Present

Seoul, South Korea

- Developed and managed live high-frequency ETF arbitrage strategies in production for Korean markets, spanning short-horizon signal modelling through to low-latency execution.
- Developed reinforcement-learning-based order execution algorithms, designing reward shaping and state representations to optimize market impact under partial observability.
- Built end-to-end systematic trading algorithms bridging deep RL research and production deployment, with emphasis on sample efficiency and robustness to regime shifts.

Publications

Preparing Ground and Excited States Using Adiabatic CoVaR

2024 – 2025

First-author, with Prof. Bálint Koczor (University of Oxford)

Oxford, UK

- Developed a hybrid quantum-classical algorithm for preparing ground and excited eigenstates of target Hamiltonians, addressing the warm-start limitation of CoVaR via adiabatic Hamiltonian morphing. Published in *New Journal of Physics* (2025).

Generalized Gaussian Temporal Difference Error for Uncertainty-aware Reinforcement Learning

2024

Co-author, with S. Kim, J. Lee, N. Cho, S. Han (Qraft Technologies)

Seoul, South Korea

- Proposed a generalized-Gaussian model of temporal-difference error capturing higher-order moments (kurtosis), improving the estimation of aleatoric and epistemic uncertainty in deep reinforcement learning. arXiv:2408.02295.

Quantum-Inspired Classification via Efficient Simulation of Helstrom Measurement

2023 – 2024

First-author, with Prof. Daniel K. Park (Yonsei), I. F. Araujo, C. Blank

Seoul, South Korea

- Derived an efficient simulation of the Helstrom Quantum Centroid classifier for arbitrary data-copy counts, revealing that copy number is a hyperparameter rather than a monotonic improvement and enabling datasets larger than the underlying Hilbert space dimension. arXiv:2403.15308.

Awards & Honors

Best Student Poster Award, Asia Quantum Information Science Conference (AQIS)

2024

Fitzgerald Prize, Exeter College, University of Oxford

2021, 2024

East Scholarship, Exeter College, University of Oxford

2020–2024

First Place, South East Asia Mathematics Competition

2020

Talks & Presentations

Poster presentations: Asia Quantum Information Science Conference (AQIS); Quantum
Techniques in Machine Learning (QTML) 2024

Skills

Programming: Python (Advanced), C++ (Intermediate), Rust (Intermediate), L^AT_EX

Areas: Quantitative Research, Machine Learning / Reinforcement Learning, Quantum Computing

Languages: English (Fluent), Korean (Native)